

Sobol sensitivity regime map

(a) no cable

S _{wale}	$S_T=0.09$ $S_1=0.05$	$S_T=0.11$ $S_1=0.05$ <i>w. $r = E_2/E_1$</i>	$S_T=0.11$ $S_1=0.05$ <i>w. $r = E_2/E_1$</i>	$S_T=0.60$ $S_1=0.00$ <i>w. θ_{knit}</i>	$S_T=0.02$ $S_1=0.01$	$S_T=0.02$ $S_1=0.01$	$S_T=0.03$ $S_1=0.01$
	<i>W. S_{wale}</i>	<i>W. S_{wale}</i>	<i>W. S_{wale}</i>	<i>W. ν_{12}</i>	<i>w. $r = E_2/E_1$</i>	<i>w. $r = E_2/E_1$</i>	<i>w. $r = E_2/E_1$</i>
S _{course}	$S_T=0.27$ $S_1=0.20$ <i>W. S_{wale}</i>	$S_T=0.32$ $S_1=0.22$ <i>W. S_{wale}</i>	$S_T=0.32$ $S_1=0.23$ <i>W. S_{wale}</i>	$S_T=0.39$ $S_1=0.00$ <i>W. ν_{12}</i>	$S_T=0.52$ $S_1=0.32$ <i>w. $r = E_2/E_1$</i>	$S_T=0.53$ $S_1=0.33$ <i>w. $r = E_2/E_1$</i>	$S_T=0.52$ $S_1=0.34$ <i>w. $r = E_2/E_1$</i>
θ_{knit} (°)	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.99$ $S_1=0.12$ <i>W. S_{wale}</i>	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$
p (Pa)	$S_T=0.20$ $S_1=0.14$ <i>W. S_{wale}</i>	$S_T=0.20$ $S_1=0.14$	$S_T=0.20$ $S_1=0.14$ <i>W. S_{wale}</i>	$S_T=0.10$ $S_1=0.00$ <i>W. S_{wale}</i>	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$
E ₁ (N/m) (wale)	$S_T=0.50$ $S_1=0.42$ <i>W. S_{wale}</i>	$S_T=0.44$ $S_1=0.39$	$S_T=0.44$ $S_1=0.39$	$S_T=0.22$ $S_1=0.00$ <i>W. θ_{knit}</i>	$S_T=0.44$ $S_1=0.28$ <i>W. S_{course}</i>	$S_T=0.44$ $S_1=0.28$ <i>W. S_{course}</i>	$S_T=0.45$ $S_1=0.31$ <i>W. S_{course}</i>
$r = E_2/E_1$	$S_T=0.06$ $S_1=0.03$	$S_T=0.06$ $S_1=0.03$	$S_T=0.06$ $S_1=0.03$	$S_T=0.12$ $S_1=0.00$ <i>W. S_{wale}</i>	$S_T=0.23$ $S_1=0.12$ <i>W. S_{course}</i>	$S_T=0.23$ $S_1=0.12$ <i>W. S_{course}</i>	$S_T=0.18$ $S_1=0.09$ <i>W. S_{course}</i>
ν_{12}	$S_T=0.02$ $S_1=0.01$	$S_T=0.02$ $S_1=0.01$	$S_T=0.02$ $S_1=0.02$	$S_T=0.01$ $S_1=0.00$	$S_T=0.03$ $S_1=0.02$	$S_T=0.03$ $S_1=0.02$	$S_T=0.04$ $S_1=0.02$

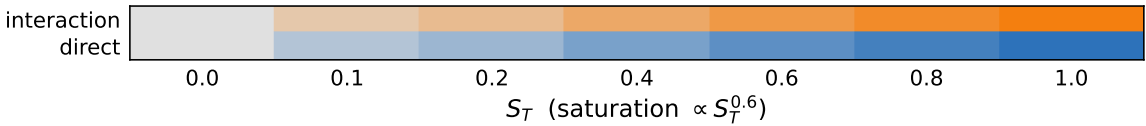
h_{crown} (m) $\tilde{H}_{x=0}$ (m⁻¹) $\tilde{H}_{y=0}$ (m⁻¹) ΔH σ_{max} (N/m) $\tilde{\sigma}$ (N/m) $\tilde{R}_{\text{bdry}}$ (N)

(b) cable

S _{wale}	$S_T=0.09$ $S_1=0.03$ <i>W. S_{course}</i>	$S_T=0.23$ $S_1=0.06$ <i>W. S_{course}</i>	$S_T=0.31$ $S_1=0.04$ <i>W. S_{course}</i>	$S_T=0.09$ $S_1=0.00$ <i>W. θ_{knit}</i>	$S_T=0.02$ $S_1=0.02$	$S_T=0.02$ $S_1=0.02$	$S_T=0.03$ $S_1=0.03$	$S_T=0.04$ $S_1=0.03$	$S_T=0.04$ $S_1=0.03$
	<i>W. p</i>	<i>W. S_{wale}</i>	<i>W. θ_{knit}</i>	<i>W. L_{rest}^{wale}</i>	<i>W. E₁</i>	<i>W. E₁</i>	<i>W. E₁</i>		
S _{course}	$S_T=0.25$ $S_1=0.17$ <i>W. p</i>	$S_T=0.48$ $S_1=0.26$ <i>W. S_{wale}</i>	$S_T=0.49$ $S_1=0.22$ <i>W. θ_{knit}</i>	$S_T=0.18$ $S_1=0.03$ <i>W. L_{rest}^{wale}</i>	$S_T=0.42$ $S_1=0.27$ <i>W. E₁</i>	$S_T=0.53$ $S_1=0.34$ <i>W. E₁</i>	$S_T=0.53$ $S_1=0.36$ <i>W. E₁</i>	$S_T=0.09$ $S_1=0.07$	$S_T=0.10$ $S_1=0.08$
θ_{knit} (°)	$S_T=0.00$ $S_1=0.00$	$S_T=0.02$ $S_1=0.00$	$S_T=0.03$ $S_1=0.00$	$S_T=0.18$ $S_1=0.06$ <i>W. S_{wale}</i>	$S_T=0.01$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.02$ $S_1=0.01$	$S_T=0.02$ $S_1=0.01$
p (Pa)	$S_T=0.22$ $S_1=0.13$ <i>W. S_{course}</i>	$S_T=0.13$ $S_1=0.04$	$S_T=0.23$ $S_1=0.05$	$S_T=0.09$ $S_1=0.01$ <i>W. L_{rest}^{wale}</i>	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.01$ $S_1=0.00$	$S_T=0.01$ $S_1=0.01$
L _{rest} ^{wale}	$S_T=0.00$ $S_1=0.00$	$S_T=0.23$ $S_1=0.02$ <i>W. E₁</i>	$S_T=0.00$ $S_1=0.00$	$S_T=0.62$ $S_1=0.35$ <i>W. S_{course}</i>	$S_T=0.02$ $S_1=0.01$	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.44$ $S_1=0.42$	
L _{rest} ^{course}	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$	$S_T=0.13$ $S_1=0.00$ <i>W. E₁</i>	$S_T=0.28$ $S_1=0.14$ <i>W. S_{course}</i>	$S_T=0.02$ $S_1=0.01$	$S_T=0.00$ $S_1=0.00$	$S_T=0.00$ $S_1=0.00$		$S_T=0.42$ $S_1=0.42$
E ₁ (N/m) (wale)	$S_T=0.52$ $S_1=0.43$ <i>W. S_{course}</i>	$S_T=0.31$ $S_1=0.14$ <i>W. L_{rest}^{wale}</i>	$S_T=0.43$ $S_1=0.17$ <i>W. L_{rest}^{course}</i>	$S_T=0.16$ $S_1=0.01$ <i>W. θ_{knit}</i>	$S_T=0.50$ $S_1=0.34$ <i>W. S_{course}</i>	$S_T=0.46$ $S_1=0.28$ <i>W. S_{course}</i>	$S_T=0.46$ $S_1=0.28$ <i>W. S_{course}</i>	$S_T=0.39$ $S_1=0.38$	$S_T=0.39$ $S_1=0.38$
$r = E_2/E_1$	$S_T=0.07$ $S_1=0.04$	$S_T=0.05$ $S_1=0.02$	$S_T=0.06$ $S_1=0.03$	$S_T=0.03$ $S_1=0.02$	$S_T=0.26$ $S_1=0.14$ <i>W. E₁</i>	$S_T=0.24$ $S_1=0.12$ <i>W. E₁</i>	$S_T=0.19$ $S_1=0.09$ <i>W. E₁</i>	$S_T=0.05$ $S_1=0.05$	$S_T=0.05$ $S_1=0.05$
ν_{12}	$S_T=0.02$ $S_1=0.01$	$S_T=0.03$ $S_1=0.00$	$S_T=0.08$ $S_1=0.00$ <i>W. S_{course}</i>	$S_T=0.03$ $S_1=0.01$	$S_T=0.02$ $S_1=0.01$	$S_T=0.03$ $S_1=0.01$	$S_T=0.04$ $S_1=0.02$	$S_T=0.01$ $S_1=0.01$	$S_T=0.00$ $S_1=0.00$

h_{crown} (m) $\tilde{H}_{x=0}$ (m⁻¹) $\tilde{H}_{y=0}$ (m⁻¹) ΔH σ_{max} (N/m) $\tilde{\sigma}$ (N/m) $\tilde{R}_{\text{bdry}}$ (N) $\log(1+T_{\text{wale}})$ $\log(1+T_{\text{course}})$

Direct (S₁/S_T large) Interaction (S₁/S_T small) Negligible (S_T → 0)



Hue = direct share S₁/S_T (blue → orange), saturation = S_T^{0.6} over grey; italic = dominant S₂ partner